2 Introduction to 8085 Microprocessor

The 8085 is an 8 bit microprocessor (MP) used in many applications. It is a single chip NMOS device with 6200 transistors packed on a 40 pin dual in line (DIL) package.

Features of 8085 Microprocessor

- It is an 8 bit MP that operates on a +5 V single power supply with a frequency of 3 MHz.
- It has a 16 bit address line. So, can access 64 (2^{16}) KB of memory.
- It gives 8 bit address to access 256 (2^8) I/O ports.
- It has an 8 bit multiplexed bus AD_0–AD_7.
- It has an 8 bit accumulator, a flag register, an instruction register, six 8 bit general purpose registers, B, C, D, E, H, and L, and two 16 bit registers.
- It has five hardware interrupts: TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR.
- It has five addressing modes: register, direct, indirect, immediate, and implied.
- It has five flags: S, Z, AC, P, and CY.
- It has serial I/O control.

Architecture of 8085 Microprocessor

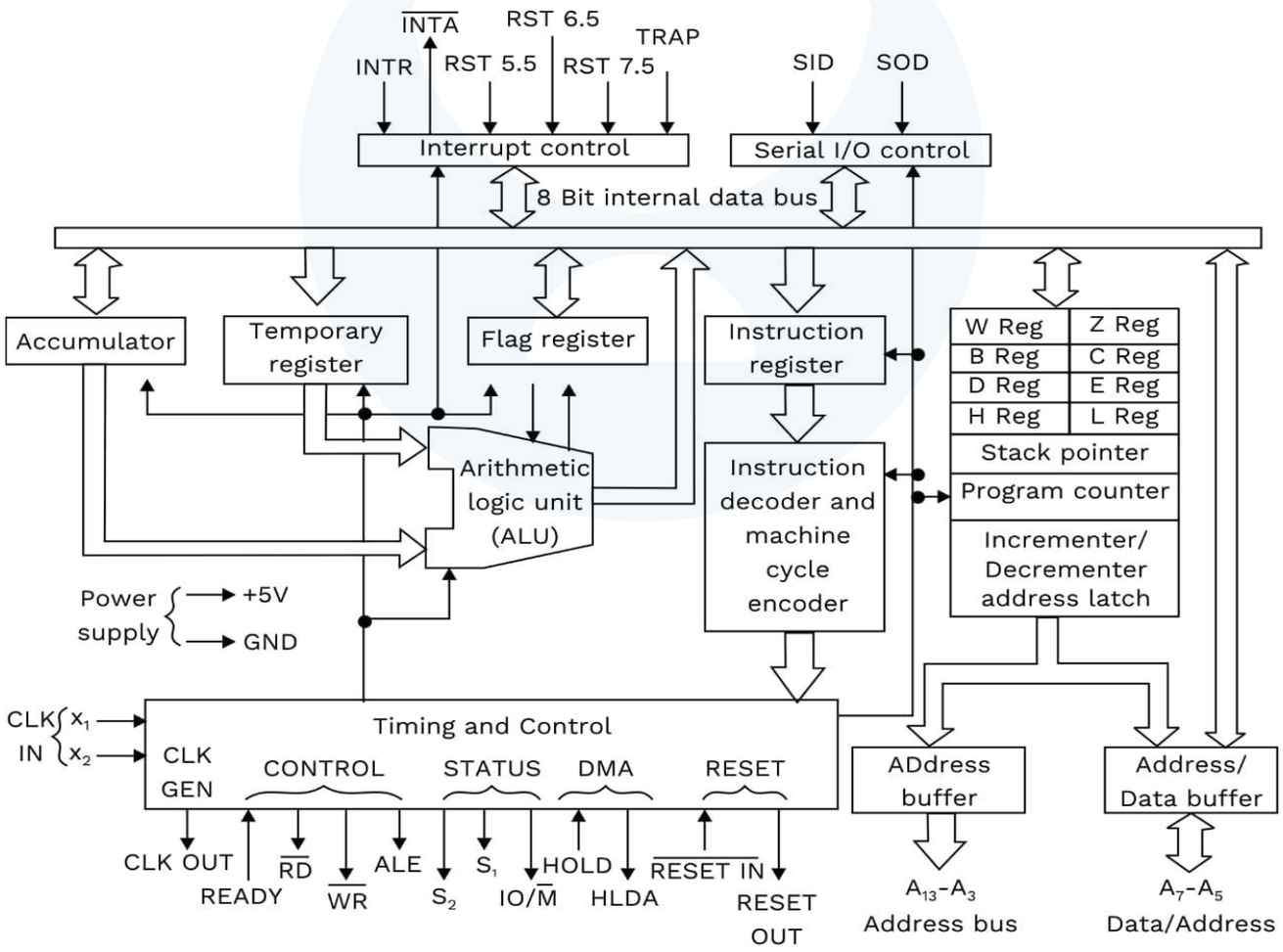


Fig. 2.1 Architecture of 8085 MP

It consists of the following functional blocks.

1. Registers
2. ALU
3. Instruction decoder and machine cycle encoder
4. Address buffer
5. Address or data buffer
6. Incrementer or decrementer address latch
7. Interrupt control
8. Serial I/O control
9. Timing and control circuitry

1. Registers

a. General purpose registers: B, C, D, E, H, and L (8 bit)

i. These registers can be used as separate 8 bit registers, as well as, in pairs. Three register pairs, each 16 bit are BC, DE, and HL (higher byte in the first register and lower byte in the second register).

ii. For example, if 2022 is stored in register pair DE, 22 will be loaded in D and 20 will be stored in E.

iii. HL pair also functions on a data pointer or memory pointer.

b. Temporary registers: temporary data registers and W and Z registers. These registers are not accessible to the programmer; they are used internally by the processor.

c. Special purpose registers: register A (accumulator), flag register, and instruction register.

i. Register A: it is an 8 bit register used in arithmetic logic, load, store, and I/O operations.

ii. Flag register: it is an 8 bit register having five flags.

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
S	Z		AC		P		CY

Fig. 2.2 Flag Register

iii. S: (sign flag) after the execution of arithmetic or logical operation, if bit D_7 is 1 (S flag is set), it means the result is a negative number and if bit D_7 is 0, it means the result is a positive number.

iv. Z: (zero flag) flag sets if the result in ALU is zero and resets if the result is non-zero.

v. AC: (auxiliary carry) flag sets if there is a carry from lower nibble (D_3) to higher nibble (D_4).

vi. P: (parity flag) flag sets if there is an even number of 1s in the result in accumulator and resets if there is odd number of 1s in the result.

vii. CY: (carry flag) flag sets if there is an overflow from bit D_7 otherwise resets. It also serves as a borrow flag for subtraction.

viii. Instruction register (IR): it stores the opcode of the instruction, which the processor fetches from the memory.

d. 16 bit registers: program counter (PC) and stack pointer (SP).

i. PC: it stores the address of the next instruction to be fetched.

ii. SP: it is used to store temporary information. It holds the address of the last stack entry.

2. ALU: all arithmetic (addition and subtraction) and logical (complement, AND, OR, and XOR) operations are performed in this unit.

3. Instruction decoder: the MP first fetches the opcode of the instruction from memory and stores it in the IR. It is, then, sent to the instruction decoder, which decodes it and accordingly gives the timing and control signals.

4. Address buffer: it is an 8 bit unidirectional buffer, which drives higher order address bus A_{15}–A_8.

5. Address or data buffer: it is an 8 bit bidirectional buffer, which drives the multiplexed address or data bus (AD_7–AD_0).

6. Incrementer or decrementer address latch: it increments or decrements the contents of PC or SP during the execution of instructions.

7. Interrupt control: it accepts the five hardware interrupts and acknowledges them.

8. **Serial I/O control:** it controls the serial communication of data through serial input data (SID) and serial output data (SOD) pins.

9. **Timing and control unit:** this unit provides the required timing and control signals for fetching, loading operations, and execution of instructions and interfacing external devices with the processor.

Pin Diagram of 8085 Microprocessor

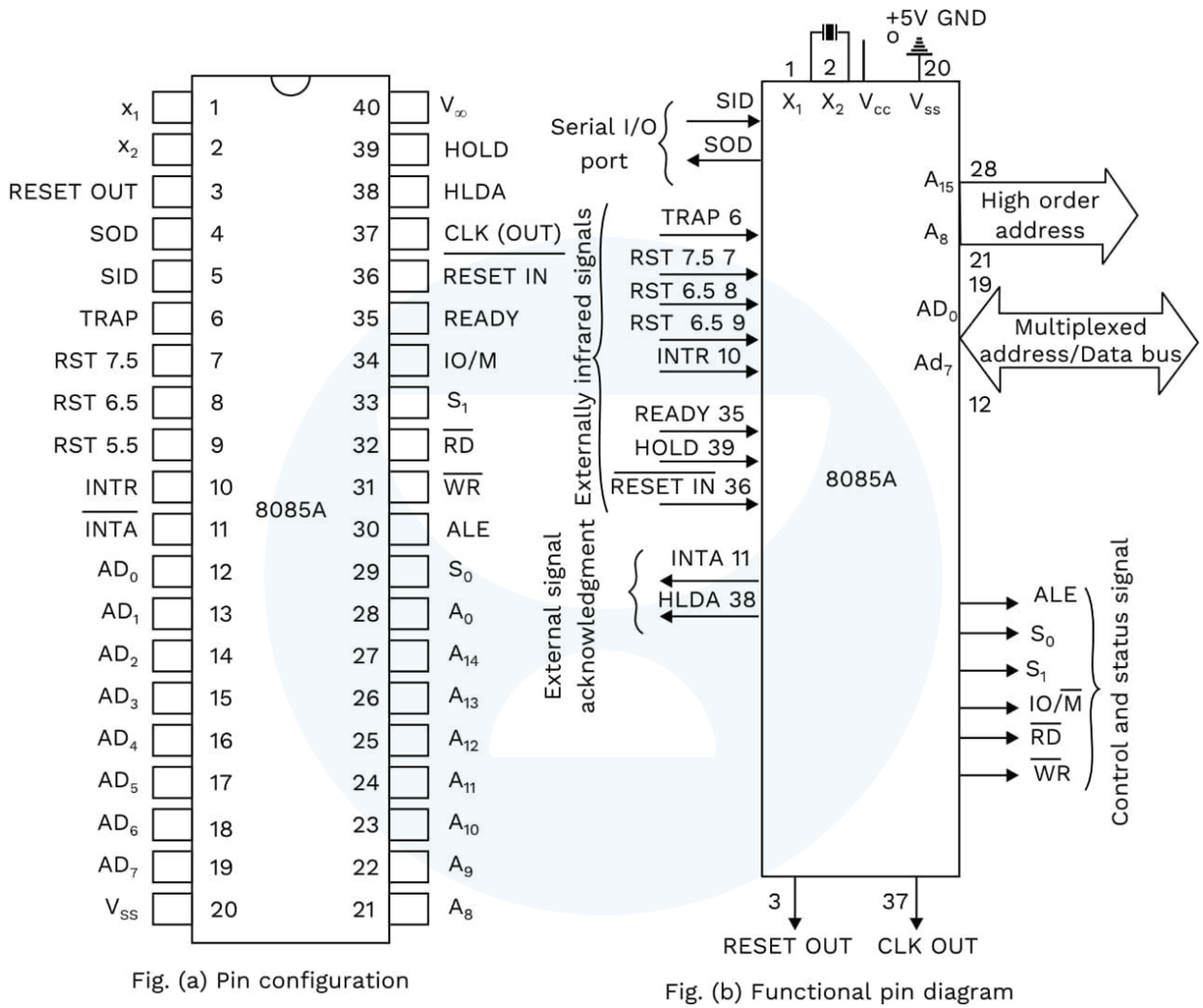


Fig. (a) Pin configuration

Fig. (b) Functional pin diagram

Fig. 2.3 Pin Diagram of 8085

According to the function, signals of 8085 are classified as follows.

Power Supply and Frequency Signals

X_1 and X_2: a crystal oscillator is connected to these pins having oscillation frequency of 6 MHz as operating frequency of 8085 is 3 MHz.

V_{cc}: it is connected to +5 V power supply.

V_{ss}: it is connected to the ground.

Address Bus and Data Bus

AD_7–AD_0: 8 bit bidirectional multiplexed address or data bus; during the first part of the machine cycle memory or I/O, the address appears on this bus and during the rest of the part of machine cycle, it acts as a data bus.

A_{15}–A_8: upper half of the 16 bit address appears on this bus.

Control and Status Signals

- *RD* and *WR*: these are active low control signals. *RD* signal indicates that the selected I/O or memory device is to be read and data are available on the data bus. *WR* signal indicates that the data on the data bus is to be written into a selected memory or I/O location.
- IO/ *M*, S_0 and S_1: these are status signals. IO/ M indicates whether I/O operation or memory operation is carried out. S_0 and S_1 indicate the type of machine cycle in progress.
- Address latch enable (ALE): it is a special signal. The latching of the lower half of the address bus is done by using this signal. It goes high during the first clock cycle of the machine cycle.
- Interrupt signals: 8085 has five hardware interrupt signals TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR. The MP recognises these interrupts at the end of the current instruction. RST 7.5, RST 6.5, and RST 5.5 are restart interrupts. They are maskable. TRAP has the highest priority among the interrupts and it is non-maskable. INTR has the lowest priority.

Serial I/O signals

- Serial input data (SID): it is an input signal used to accept data serially bit by bit from external device.
- Serial output data (SOD): it is an output signal used to transmit data serially bit by bit to external device.

Direct Memory Access (DMA) Signals

- HOLD: this signal indicates that another device is requesting the use of the address, data, and control bus.
- HLDA: this active high signal is used to acknowledge the HOLD request.

Reset Signals

- Reset in: it resets the PC to zero. It also resets interrupt enable and HLDA flip-flops. It does not affect any flag or register except the instruction register.
- Reset out: it indicates that the processor is being reset.

READY: it is a signal sent by an external I/O device to the MP. When READY is high, it indicates the I/O device is ready to send or receive data. When READY is low, the MP waits till READY becomes high.

Instruction cycle: in a program, instructions are executed one by one in sequence. The fetching, decoding, and execution of each instruction have an instruction cycle that consists of a few read–write operations between processor and memory or I/O devices. Each memory or I/O operation takes a time period called machine cycle. Each machine cycle consists of a few clock cycles referred to as T-states. It means each instruction cycle consists of machine cycles and machine cycles consist of T-states.

Important Machine Cycles of 8085

Following are the important machine cycles of 8085.

1. **Opcode fetch cycle**: the first machine cycle of each instruction is opcode fetch cycle, in which the processor fetches the opcode of instruction from the memory. This machine cycle takes four T-states. Each T-state starts with the falling edge of the clock cycle.
   a. T_1 state: during T_1 state, higher order bits of PC are loaded on A_{15}–A_8 and lower order bits on AD_7–AD_0 lines. MP activates ALE signal to latch lower order bits of the address to the internal latch before it disappears. During this cycle status signals IO/ *M* = 0, S_0 = 1 and S_1 = 1.
   b. T_2 state: during T_2 state, lower order address disappears from AD_7–AD_0 lines, *RD* goes low to enable the addressed memory location. The contents of the addressed memory location are placed on AD_7–AD_0.

c. T_3 state: during T_3 state, data from the data bus are loaded into the instruction register (IR) and *RD* goes high.

d. T_4 state: in this state, opcode is decoded and then encoded into various machine cycles to complete the task of instruction.

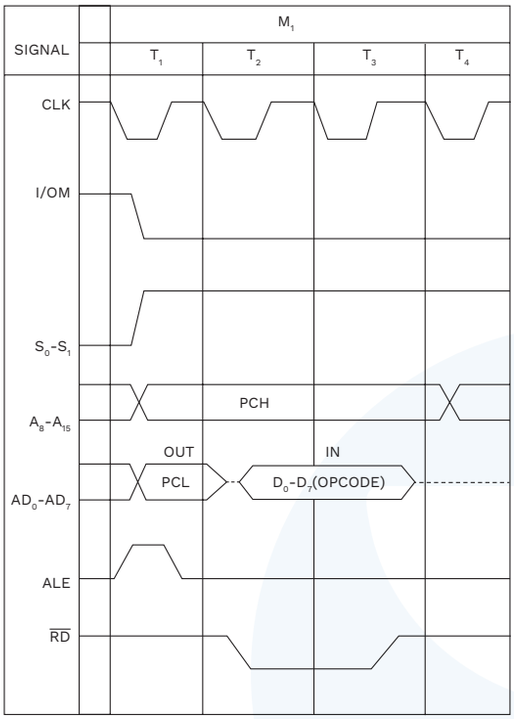


Fig. 2.4 Opcode Fetch Cycle

2. Memory read cycle: in this cycle, MP reads the contents of a memory location. There are three T-states in this cycle.

a. T_1 state: during T_1 state, MP places the address on the address lines from SP, PC, or general register pair and activates ALE signal to latch lower order bits of the address. During this cycle, status signals IO/ *M* = 0, S_0 = 1, and S_1 = 0.

b. T_2 state: during T_2 state *RD* goes low to enable the addressed memory location. The contents of addressed memory locations are placed on AD_7–AD_0.

c. T_3 state: during T_3 state, data from the data bus are loaded into the specified register and *RD* goes high.

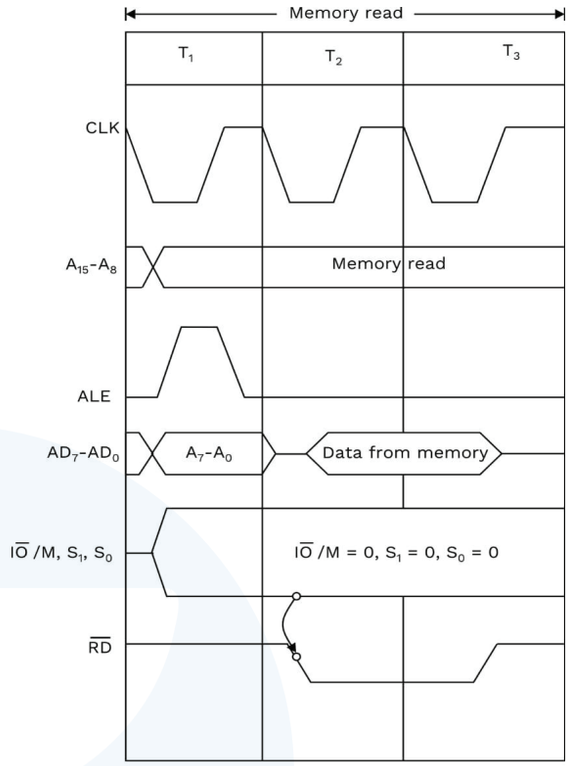


Fig. 2.5 Memory Read Cycle

3. Memory write cycle: in this cycle, MP writes the data into the memory. It takes three T-states.

a. T_1 state: during T_1 state, MP places the address on the address lines from SP, PC, or general register pair and activates ALE signal to latch lower order bits of the address. During this cycle, status signals IO/ *M* = 0, S_0 = 0 and S_1 = 1.

b. T_2 state: during T_2 state, MP places the data on the data bus and sends *WR* low for writing data into specified memory locations.

c. T_3 state: during T_3 state, *WR* goes high to terminate the write operation.

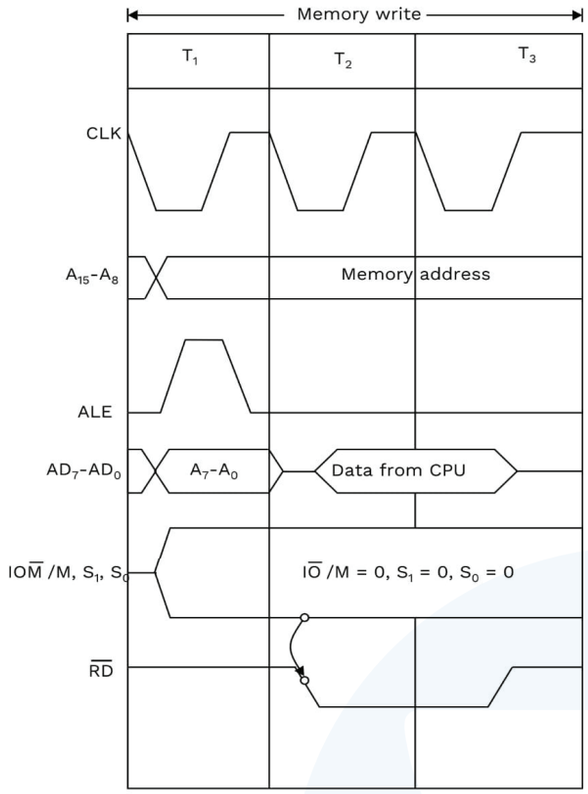


Fig. 2.6 Memory Write Cycle

4. **I/O read and write cycles:** they are the same as memory read and write cycles, except for the status signal IO/ *M*. It is high (1) to indicate I/O operation.

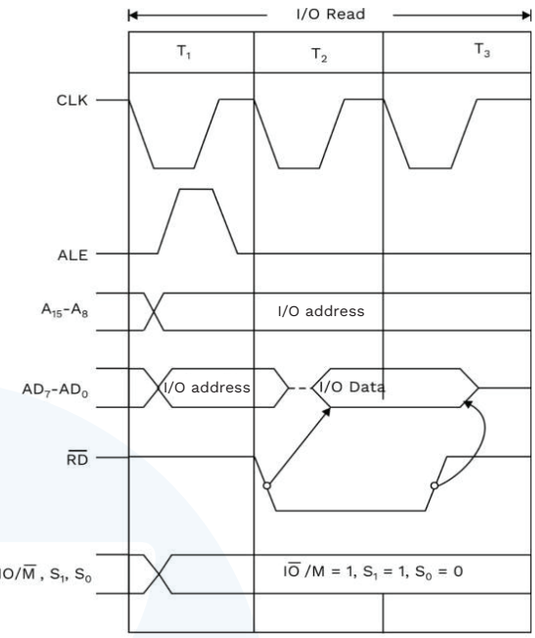


Fig. 2.7 I/O Read Cycle

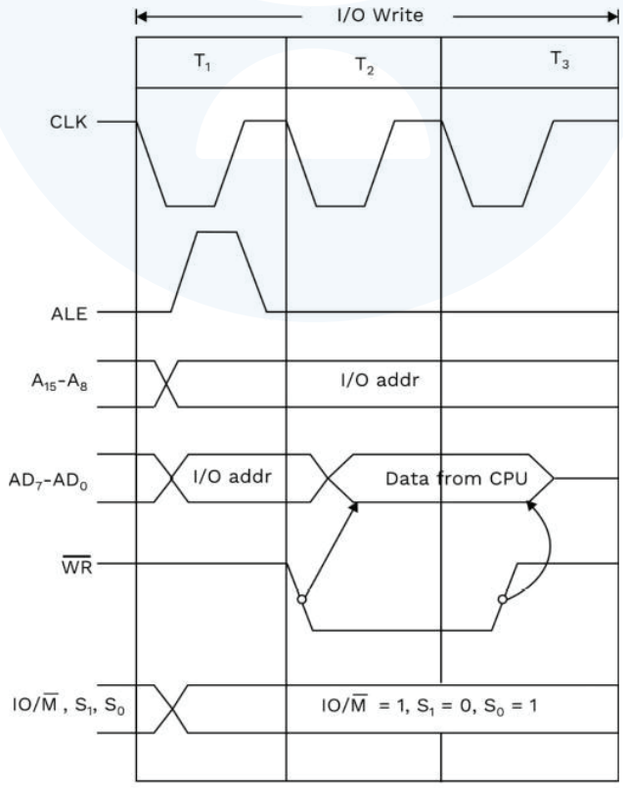


Fig. 2.8 I/O Write Cycle

Practice Exercise

Level I

1. Which one of the following signal combinations will generate $\overline{MEMR}$ signal? **[ESE-EE,00]**

(A) $IO/\overline{M} \wedge \overline{RD}$
(B) $IO/\overline{M} \vee \overline{RD}$
(C) $\overline{IO/\overline{M}} \vee \overline{RD}$
(D) $\overline{IO/\overline{M}} \wedge \overline{RD}$

2. Which one of the following sets of status signals is used in an 8085 microprocessor to uniquely identify the seven machine cycles? **[ESE-EE,00]**

(A) $IO/\overline{M}, \overline{RD}, \overline{WR}$
(B) $S_1, S_0, \overline{INTA}$
(C) $IO/\overline{M}, S_1, S_0$
(D) S_2, S_1, S_0

3. The number of output pins of an 8085 microprocessor is: **[ESE-EE,02]**

(A) 40 (C) 21
(B) 27 (D) 19

4. Which of the following does not take place when an 8085 processor is reset? **[ESE-EE,05]**

(A) 8085 gives reset out signal to reset external hardware.
(B) 8085 resets program counter to FFFF H.
(C) The interrupt system is disabled.
(D) The buses are tristated.

5. In a microcomputer, why wait states are used? **[ESE-E&T,07]**

(A) To make the processor wait during a DMA operation
(B) To make the processor wait during an interrupt processing
(C) To make the processor wait during a power shutdown
(D) To interface slow peripherals to the processor

6. Which signal of 8085 microprocessor is used to insert wait states? **[ESE-EE,07]**

(A) READY (C) HOLD
(B) ALE (D) INTR

7. If the status of the control lines S_1 and S_0 is low, then a 8085 microprocessor performs: **[ESE-EE]**

(A) reset operation
(B) HOLD operation
(C) halt operation
(D) interrupt acknowledge

8. Which one of the following statements is correct? **[ESE-07]**
The ALE line of intel 8085 microprocessor is used to:

(A) latch the output of an I/O instruction into an external latch.
(B) deactivate the chip-select signal from memory devices.
(C) latch the 8 bits of address lines AD7–AD0 into an external latch.
(D) find the interrupt enable status of the TRAP interrupt.

9. Which one of the following statements is correct? **[ESE-E&T,10]**
In the 8085 microprocessor, the READY signal is useful when the CPU communicates with a:

(A) a PPI chip
(B) a DMA controller chip
(C) a slow peripheral device
(D) a fast peripheral device

10. Which one of the following keeps track of the instruction execution sequence? **[ESE-E&T-91]**

(A) Accumulator
(B) Program counter
(C) Stack pointer
(D) Instruction register

11. Which one of the following flags is not used for branching in a microprocessor? **[ESE-E&T-91]**

(A) Carry flag
(B) Auxiliary carry flag
(C) Overflow flag
(D) Parity flag

12. In the 8085 microprocessor, the value of the most significant bit of the result following the execution of any arithmetic or Boolean instruction is stored in the: **[ESE-E&T-98]**

(A) carry status flag
(B) auxiliary carry flag
(C) sign status flag
(D) zero status flag

13. A microprocessor is ALU: **[ESE-EE,03]**

(A) and control unit on a single chip.
(B) and memory on a single chip.
(C) register unit and I/O device on a single chip.
(D) register unit and control unit on a single chip.

14. The program counter in an 8085 microprocessor is a 16 bit register because: **[ESE-EE-03]**

(A) it counts 16 bits at a time.
(B) there are 16 address lines.
(C) it facilitates the user storing 16 bit data temporarily.
(D) it has to fetch two 8 bit data at a time.

15. Processor status word of 8085 microprocessor has five flags. Which are these five flags? **[ESE-EE-06]**

(A) S, Z, AC, P, and CY
(B) S, OV, AC, P, and CY
(C) S, Z, OV, P, and CY
(D) S, Z, AC, P, and OV

16. The 8085 μP has two registers known as primary data pointers. These are the registers: **[ESE-E&T,10]**

(A) B and C
(B) D and E
(C) H and L
(D) C and D

17. In an 8085 microprocessor, HL register pair is used for storing: **[ESE-98]**

(A) address of memory
(B) data
(C) address of next instruction
(D) current instruction

18. The highest integer that can be stored in an 8 bit accumulator is: **[KAS,04]**

(A) 64
(B) 144
(C) 256
(D) 255

19. How many general purpose 8 bit registers are there in an 8085 microprocessor? **[ESE-05]**

(A) 3
(B) 6
(C) 9
(D) 12

20. Which of the following flag conditions are not available in an 8085 processor? **[UTPCS,05]**

(A) Zero flag
(B) Parity flag
(C) Overflow flag
(D) Auxiliary carry flag

21. Which one of the following statements does not describe the property or characteristic of a stack pointer register in an 8085 microprocessor? **[ESE-EE,07]**

(A) It points to the top of the stack.
(B) It is up or down counter.
(C) It is automatically initialised to 0000H.
(D) It is a 16 bit register.

22. In a microprocessor, the program counter points to the address location from where the: **[BSNL-JTO-TC,09]**

(A) current byte is to be fetched
(B) next byte is to be fetched
(C) next byte is to be stored
(D) current byte is to be added

23. In an 8085 microprocessor, the accumulator is a: **[DRDO-E&C,08]**

(A) 4 bit register
(B) 8 bit register
(C) 16 bit register
(D) 32 bit register

24. ALU carries out: **[MPPCS-EE,09]**
(A) arithmetic operation
(B) logic operation
(C) both arithmetic and logic operation
(D) None of the above

25. In 8085 pairing of registers B, C, D, E, H, and L is: **[MPPCS-EE,09]**
(A) B-C, D-E, H-L
(B) B-D, C-E, H-L
(C) B-C, D-L, H-E
(D) Either of these

26. In 8085 μP: **[MPPCS-EE,09]**
(A) P flag is reset when the result has even parity.
(B) P flag is reset when the result has odd parity.
(C) P flag is set when the result has odd parity.
(D) Any of the above

27. Memory-mapped I/O scheme for the allocation of address to memories and I/O devices is used for: **[ESE-EE,02]**
(A) small systems
(B) large systems
(C) both large and small systems
(D) very large systems

28. An 8085 μP uses a crystal of frequency 6.25 MHz. The time of one T state will be: **[UPPCS,03]**
(A) 320 ns
(B) 640 ns
(C) 960 ns
(D) 1280 ns

29. The duration of one T-state in an 8085 microprocessor that uses a crystal of 5.00 MHz is: **[BSNL-JTO-EE,09]**
(A) 0.2 μs
(B) 0.4 μs
(C) 2.5 μs
(D) 5.0 μs

30. A microprocessor using a 3 MHz clock has three 'T' states in each machine cycle. If an instruction cycle needs four machine cycles, how much time will be taken to complete the execution of this instruction? **[ESE-EE,05]**
(A) 333 ns
(B) 1333 ns
(C) 1 μs
(D) 4 μs

31. An instruction cycle is the time in which a hard-wired control completes four functions. What is the correct sequence of these functions? **[ESE-05]**
(A) Fetch-update-decode-execute
(B) Fetch-decode-update-execute
(C) Decode-execute-fetch-update
(D) Execute-decode-fetch-update

32. In an 8085 microprocessor, the I/O devices can be used in: **[UPPSC,20]**
(A) memory mapped I/O only
(B) I/O mapped I/O only
(C) memory mapped I/O or I/O mapped I/O
(D) None of the above

Level II

1. Some of the pins of an 8085 CPU and their functions are listed below. Identify the answer that matches the pins to their respective functions. **[GATE-IN-03]**

P.	RST 7.5	I. Selects IO or memory
Q.	HOLD	II. Demultiplexes the address == bus
R.	$IO/\overline{M}$	III. Is a vectored interrupt
S.	ALE	IV. Facilitates direct memory as
		V. Is a clock
		VI. Selects BCD mode of operation

(A) P-III, Q-II, R-I, S-IV
(B) P-IV, Q-I, R-V, S-III
(C) P-III, Q-IV, R-I, S-II
(D) P-II, Q-III, R-VI, S-I

2. The frequency of the driving network connected between pins 1 and 2 of an 8085 ==== be: **[ESE]**

(A) equal to the desired clock frequency.
(B) twice the desired clock frequency.
(C) four times the desired clock frequency
(D) eight times the desired clock frequency.

3. **Assertion (A):** The data bus and address bus of an 8085 microprocessor are multipliers.

Reason (R): Multiplexing reduces the number of pins. **[ESE]**

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

4. Consider the following statements.

I. Bus is a group of wires carrying information.
II. Bus is needed to achieve reasonable speed of operation.
III. Bus can carry data or address.
IV. A bus can be shared by more than one device.

Select the correct answer from the codes given below. **[ESE-E&T,09]**

(A) I only
(B) I and II only
(C) II, III, and IV only
(D) I, II, III, and IV

5. Consider the following statements.

I. In an 8085 microprocessor, data bus and address bus are multiplexed in order to increase the speed of the microprocessor.
II. In an 8085 microprocessor, data bus and address bus are multiplexed in order to reduce the number of pins.
III. In an 8085 microprocessor, data bus and address bus are multiplexed in order to connect more peripheral chips.

Which of the above statements is/are correct? **[ESE-EE]**

(A) I only
(B) II only
(C) II and III
(D) I, II, and III

6. Match List-I (microprocessor pin) with List-II (signals on pin) and select the correct answer using the codes given below the lists. **[ESE-EE]**

List-I

a. TRAP
b. HLDA
c. RESET
d. ALE

List-II

I. Interrupt
II. Initialising
III. Enable
IV. Memory access

Codes:

	a	b	c	d
(A)	I	II	IV	III
(B)	III	II	IV	I
(C)	I	IV	II	III
(D)	III	IV	II	I

7. Match List-I (instruction of 8085) with List-II (function of instruction) and select the correct answer using the codes given below the lists. **[ESE-E&T]**

List-I

a. RST 6.5
b. INTR
c. TRAP
d. HLDA

List-II

I. Highest priority of interrupts
II. Vector interrupt
III. Interrupt request
IV. HOLD acknowledge

Codes:

	a	b	c	d
(A)	II	IV	I	III
(B)	I	III	II	IV
(C)	II	III	I	IV
(D)	I	IV	II	III

8. Match List-I with List-II and select the correct answer using the codes given below the lists. **[ESE-07]**

List-I

a. SID, SOD
b. READY
c. TRAP
d. ALE

List-II

I. Wait state
II. Interrupt
III. Serial data transfer
IV. Memory of I/O read/write
V. Address latch control

Codes:

	a	b	c	d
(A)	III	I	V	II
(B)	III	I	II	V
(C)	IV	III	II	V
(D)	IV	III	I	II

9. In the context of an 8085 microprocessor, which of these is the correct matching combination between column A and B. **[TC-09]**

Column A		Column B
P.	ALE	I. Rotate accumulator left
Q.	PSW	II. Compare with accumulator
R.	CMA	III. Program status word
S.	RLC	IV. Address latch word
		V. Program stack word
		VI. Arithmetic logic enabled
		VII. Complement accumulator
		VIII. Rotate accumulator left through carry

(A) P-VI, Q-V, R-II, S-VIII
(B) P-IV, Q-III, R-II, S-VIII
(C) P-IV, Q-III, R-VII, S-I
(D) P-VI, Q-V, R-VII, S-I

10. An m-bit microprocessor has an m-bit: **[GATE-IN,01]**

(A) flag register
(B) instruction register
(C) data counter
(D) program counter

11. **Assertion (A):** A processor can reference a memory stack without specifying an address. **Reason (R):** The address is always available and automatically updated in the stack pointer. **[ESE-E&T,01]**

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

12. Which one of the following registers of an 8085 microprocessor is not a part of programming model? **[ESE-E&T,06]**

(A) Instruction register
(B) Memory address register
(C) Status register
(D) Temporary data register

13. Consider the following registers.

I. Accumulator and B register
II. B and C registers
III. D and E registers
IV. H and L registers

Which of these 8 bit registers of 8085 μP can be paired together to make a 16 bit register? **[ESE-E&T,02]**

(A) I, III, and IV
(B) II, III, and IV
(C) I and II
(D) I, II, and III

14. Consider the following statements.

I. In an 8085 microprocessor, the pairs of general purpose registers for 16 bit operation are A–B, C–D, and H–L.
II. In an 8085 microprocessor, the data and the address multiplexed bus are known as AD7–AD0.
III. In an 8085 microprocessor, ALE is used to operate multiplexed data address bus.

Which of e statements are correct? **[ESE-E&T,07]**

(A) I, II, and III
(B) II and III only
(C) I and III only
(D) I and II only

15. In an Intel 8085A, what is the content of Instruction Register (IR)? **[ESE-EE,08]**

(A) Opcode for the instruction being executed
(B) Operand for the instruction being executed
(C) Opcode for the instruction to be executed next
(D) Operand for the instruction to be executed next

16. Both ALU and control section of CPU employ which special purpose storage location? **[ESE-EE,08]**
(A) Buffers
(B) Decoders
(C) Accumulators
(D) Registers

17. The contents of the program counter (PC), when a microprocessor is reading from 2FFF H memory location will be: **[ESE-EE,08]**
(A) 2FFE H
(B) 2FFF H
(C) 3000 H
(D) 3001 H

18. The content of the program counter of an 8085 microprocessor is: **[ESE-EE,10]**
(A) the total number of instructions in the program already executed.
(B) the total number of times a subroutine is to be executed next.
(C) the memory address of the instruction that is being currently executed.
(D) the memory address of the instruction that is to be executed next.

19. **Assertion (A):** Address bus is unidirectional.
Reason (R): Data bus is bidirectional. **[ESE-EE,10]**
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

20. Which bit of the 8 bit flag register is affected when during the execution of an arithmetic operation, there is a carry out of bit 3 to bit 4? **[ESE-02, EE,1A(c)/2]**
(A) Carry flag
(B) Zero flag
(C) Sign fag
(D) Auxiliary carry flag

21. The sign flag of an 8085 microprocessor is set to 1, if: **[ESE-04, EE,1A(b)/2]**
(A) the result of an arithmetic operation is zero.
(B) the most significant bit of the result of an arithmetic or logic operation is 1.
(C) there is a carry from addition or borrow from subtraction.
(D) the result of an operation carries four 1s in accumulator.

22. Arithmetic logic unit (ALU) of an 8085 microprocessor consists of: **[ESE-EE,11]**
(A) accumulation, temporary register, arithmetic, and logic circuits.
(B) accumulator, arithmetic, logic circuits, and five flags.
(C) accumulator, arithmetic, and logic circuits.
(D) accumulator, temporary register, arithmetic, logic circuits, and five flags.

23. The description of a program counter (PC) in 8085 microprocessor is: **[ESE-EE,11]**
(A) an up/down counter.
(B) an 8 bit register.
(C) initialised automatically by a microprocessor.
(D) used to point to stack memory area.

24. Which of the following statements about program counter (PC) is/are correct?
I. It is a register.
II. It performs logical operations.
III. It is a cell in ROM.
IV. During the execution of the current instruction, its contents change.
Select the correct answer from among the codes given below. **[ESE-08]**
Codes:
(A) I only
(B) II and III
(C) I and IV
(D) III and IV

25. Consider the following.

I. Sign flag
II. Trap flag
III. Parity flag
IV. Auxiliary carry flag

Which of the above flags is/are present in an 8085 microprocessor? **[ESE-EE,09]**

(A) I only
(B) I and II
(C) II and III
(D) I, III, and IV

26. Consider the following features in an 8085 microprocessor system with memory mapped I/O.

I. I/O devices have 16 bit addresses.
II. I/O devices are accessed using 'in' and 'out' instructions.
III. There can be a maximum of 256 input devices and 256 output devices.
IV. Arithmetic and logic operations can be directly performed with the I/O data.

Select the correct answer from among the codes given below. **[ESE-E&T,03]**

Codes:

(A) I, II, and IV
(B) I, III, and IV
(C) II and III
(D) I and IV

27. Memory-mapped I/O scheme for the allocation of address to memories and I/O devices, is used for: **[ESE-EE,02]**

(A) small systems
(B) large systems
(C) both large and small systems
(D) very large systems

28. What is the total number of memory locations and input-output devices that can be addressed with a processor having 16 bits address bus, using memory mapped I/O? **[ESE-EE,04]**

(A) 64 K memory locations and 256 I/O devices
(B) 256 I/O devices and 65279 memory locations
(C) 64 K memory locations and no I/O devices
(D) 64 K memory locations or I/O devices

29. Consider the following statements.

I. Indirect addressing is not possible for I/O mapped I/O port addresses.
II. Pointers cannot be used to access memory mapped I/O addresses.
III. Fewer machine instructions can be used with I/O mapped I/O addressing as compared to memory mapped I/O addressing.
IV. With an 8085 microprocessor, one can access at the most 512 devices with unique addressing using I/O mapped I/O addressing.

Which of these statements are correct? **[ESE-EE,06]**

(A) I, II, and III
(B) II and IV
(C) III and IV
(D) I and III

30. In an 8085 microprocessor based system, the maximum number of memory and I/O devices that can be addressed to is/are: **[2/EE/1A(f)/2]**

(A) 2^8
(B) 2^{16}
(C) $2^8 + 2^{16}$
(D) Infinite

31. In connection with memory mapped I/O which one of the following statements is not true? **[BSNL-JTO-TC,09]**

(A) The processor treats an interface register as part of the memory system.
(B) It reduces the memory space available.
(C) The processor cannot manipulate I/O data residing in interface registers with the same instructions that are used to manipulate memory location.
(D) Arithmetic or logical operation can be directly performed with I/O data.

Answer Key

Level I

1. (A)	**2.** (C)	**3.** (B)	**4.** (B)	**5.** (D)	**6.** (A)	**7.** (C)	**8.** (C)	**9.** (C)	**10.** (B)
11. (B)	**12.** (C)	**13.** (D)	**14.** (B)	**15.** (A)	**16.** (C)	**17.** (A)	**18.** (D)	**19.** (B)	**20.** (C)
21. (C)	**22.** (B)	**23.** (B)	**24.** (C)	**25.** (A)	**26.** (B)	**27.** (A)	**28.** (A)	**29.** (B)	**30.** (D)
31. (A)	**32.** (C)								

Level II

1. (C)	**2.** (A)	**3.** (A)	**4.** (D)	**5.** (B)	**6.** (C)	**7.** (C)	**8.** (B)	**9.** (C)	**10.** (B)
11. (D)	**12.** (C)	**13.** (B)	**14.** (C)	**15.** (A)	**16.** (D)	**17.** (C)	**18.** (D)	**19.** (B)	**20.** (D)
21. (B)	**22.** (D)	**23.** (C)	**24.** (C)	**25.** (D)	**26.** (D)	**27.** (A)	**28.** (D)	**29.** (D)	**30.** (B)
31. (C)									